

1. Abstract

This project explores and implements a dual-phase automated pipeline for object detection and terrain analysis, completely bypassing the need for traditional, manually annotated bounding-box training datasets. The system employs a hybrid methodology: a Deep Learning approach using a pre-trained Convolutional Neural Network (CNN) for semantic feature extraction, and a Classical Computer Vision approach utilizing Multi-Scale Template Matching and Non-Maximum Suppression (NMS) for scale-invariant structural detection. Together, these pipelines establish a highly robust, automated toolkit capable of identifying specific geological targets within complex, unmapped environments.

2. Methodology: Target-Specific Detection Pipelines

To achieve comprehensive terrain analysis, this project utilizes a dual-methodology system. Recognizing that different geological targets possess fundamentally different visual characteristics, the pipeline intelligently routes processing based on the specific morphological properties of the target material.

2.1 Iron Oxide Detection: Semantic Feature Extraction

For the detection of Iron Oxide deposits, the system relies on high-level semantic feature extraction. Iron oxide formations present highly variable, complex textures and color signatures that are difficult to define with rigid geometry.

- A pre-trained CNN acts as a deep-feature encoder, capturing the complex visual hierarchy (spectral properties, textures, and non-linear patterns) and compressing it into a high-dimensional mathematical vector.

2.2 Layered Rock Detection: Multi-Scale Structural Matching

For the detection of Layered Rock formations, the system shifts to a classical computer vision approach. Layered rocks are defined by specific, identifiable geometric structures (parallel striations) that appear at vastly different physical scales depending on camera proximity.

- The system utilizes algorithmic structural matching across dynamically scaled image intervals to identify these precise geometric patterns, ensuring detection regardless of the formation's physical distance from the lens.
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3. Phase I: Deep Learning Pipeline (ResNet-50)

The deep learning phase utilizes a ResNet-50 architecture, initialized with ImageNet weights, functioning as a feature extractor with its final classification layer truncated.

3.1 Preprocessing and Memorization

To establish a reference database, target objects (e.g., io1.jpeg through io8.jpeg) undergo strict spatial and normalization transformations to satisfy ResNet-50 constraints (resized to 224x224 pixels, RGB channel normalization). The forward pass generates unique mathematical fingerprints—1D flattened vectors—which are serialized into a reference library.

3.2 Vector Similarity Search and Scoring:

The target environment is analyzed using a sliding window algorithm, where a small region systematically scans across the image to capture localized segments, and each cropped region is transformed into a feature vector, denoted as **A**, representing its visual characteristics. These query vectors are then compared with predefined reference vectors, **B**, corresponding to known objects or patterns, using cosine similarity, a mathematical measure that evaluates similarity based on the angle between vectors; higher cosine similarity values indicate stronger matches, thereby enabling accurate detection of objects within the scanned environment. This similarity score evaluates the geometric angle between the two multi-dimensional vectors in the latent space.

Regions yielding a similarity score above a strict heuristic threshold (e.g., 0.70) are formally flagged as positive matches.

4. Phase II: Scale-Invariant Template Matching

To address the scale rigidity of the sliding window and target the specific geometry of layered rocks, the pipeline integrates classical computer vision techniques via OpenCV.

4.1 Image Pyramids and Cross-Correlation Scoring:

To achieve scale invariance, the system utilizes an image pyramid approach by dynamically resizing the search image across multiple scales, typically ranging from 20% to 150% of the original resolution. A structural template of the target object is first defined, and at each scale level, the algorithm scans the image using a sliding window mechanism. For every position, similarity between the template and the image region is computed using Normalized Cross-Correlation (`cv2.TM_CCOEFF_NORMED`), which provides a normalized score indicating the strength of the match. Higher correlation values correspond to stronger similarities, enabling the system to accurately detect objects of varying sizes across different scales.

This algorithm evaluates exact pixel-intensity relationships, outputting a structural similarity score. By subtracting the local mean intensity, this coefficient score remains highly robust against global lighting changes and shadows, allowing for a slightly lower confidence threshold (e.g., 0.45) to successfully capture partially obscured targets.

4.2 Non-Maximum Suppression (NMS)

The multi-scale scan generates clustered bounding box proposals around physical structures. To refine this data, Non-Maximum Suppression (NMS) evaluates the Intersection over Union (IoU) of overlapping coordinates (using `cv2.groupRectangles`). Proposals sharing an IoU above a defined epsilon threshold (0.2) are consolidated into a single, clean detection boundary based on the highest localized similarity score.

5. System Synthesis and Scoring Metrics

The complete system functions as an intelligent, hybrid pipeline. The table below summarizes how each respective pipeline computes and utilizes similarity scores to finalize detections:

Detection Pipeline	Target Profile	Scoring Algorithm	Output Metric	Typical Threshold
ResNet-50 (Deep Learning)	Textures, Colors, Amorphous shapes (Iron Oxide)	Cosine Similarity	Latent vector angular proximity	> 0.70
OpenCV (Multi-Scale)	Rigid Geometry, Parallel patterns (Layered Rocks)	Normalized Cross-Correlation	Pixel-intensity correlation coefficient	> 0.45

Execution Flow:

1. Input Ingestion: Ingests high-resolution target environment imagery.
2. Target Routing: Routes targets based on visual profile to the appropriate pipeline.
3. Algorithmic Execution: Computes the localized similarity scores via tensor extraction or multi-scale cross-correlation.
4. Consolidation: Passes raw spatial coordinates through NMS to produce a finalized array of precise bounding boxes, prioritizing the highest similarity scores.

6. Practical Applications

This zero-shot detection approach provides significant value in scenarios where massive training datasets are unavailable:

- Planetary Exploration: Autonomous rovers (e.g., Mars operations) can identify specific geological formations using only a handful of reference images, prioritizing scientific targets without a continuous Earth uplink.
- Automated Resource Surveying: The architecture easily adapts for drone-based aerial surveying to map mineral distributions or structural anomalies in hazardous or inaccessible environments.

7. Key Innovations and System Strengths

The development and execution of this hybrid detection pipeline yielded highly successful outcomes, demonstrating significant strengths in automated computer vision techniques.

- Zero-Shot Operational Efficiency: The system completely bypasses the massive computational overhead, hardware requirements, and time investment typically associated with training models like YOLO or R-CNN from scratch.

- **Highly Robust Feature Extraction:** The Phase I (ResNet-50) pipeline demonstrated exceptional reliability. By operating in a high-dimensional latent space, the system successfully identified target textures despite varying lighting conditions and complex Martian terrain boundaries.
- **True Scale Invariance:** The integration of the multi-scale Image Pyramid brilliantly resolved the spatial rigidity inherent in basic sliding-window approaches. The system achieved true scale invariance, accurately identifying specific structural geometries regardless of their physical distance from the camera lens.
- **Professional-Grade Data Refinement:** The application of NMS transformed raw, clustered algorithmic proposals into highly precise, actionable data, generating exceptionally clean visual overlays that bounded the target objects without visual clutter.
- **Versatile Hybrid Architecture:** Rather than forcing a single algorithm to compromise on accuracy, the pipeline intelligently adapts to the morphology of the target. This ensures maximum detection accuracy across a diverse range of geological formations.

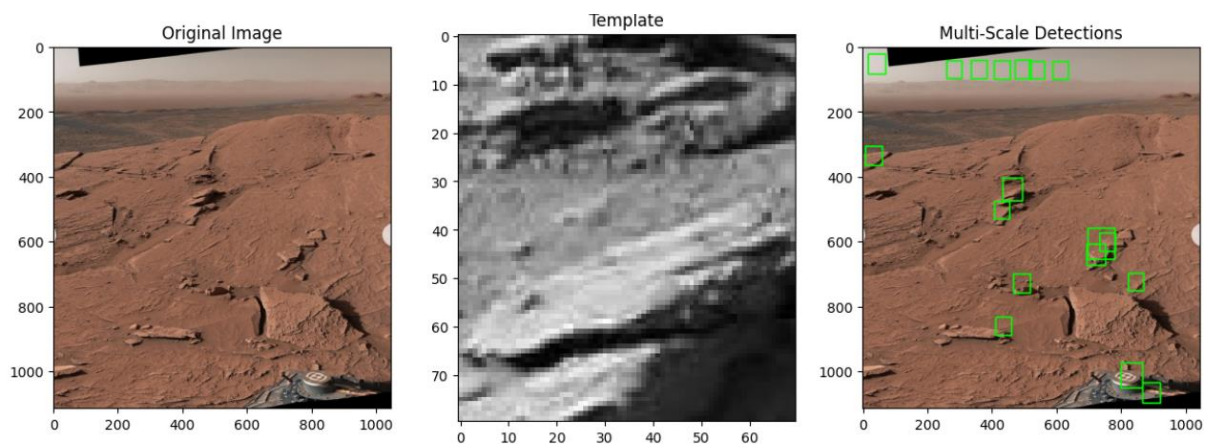


Figure 1: Multi-Scale Template Matching and Scale-Invariant Detection of Layered Rock Formations.

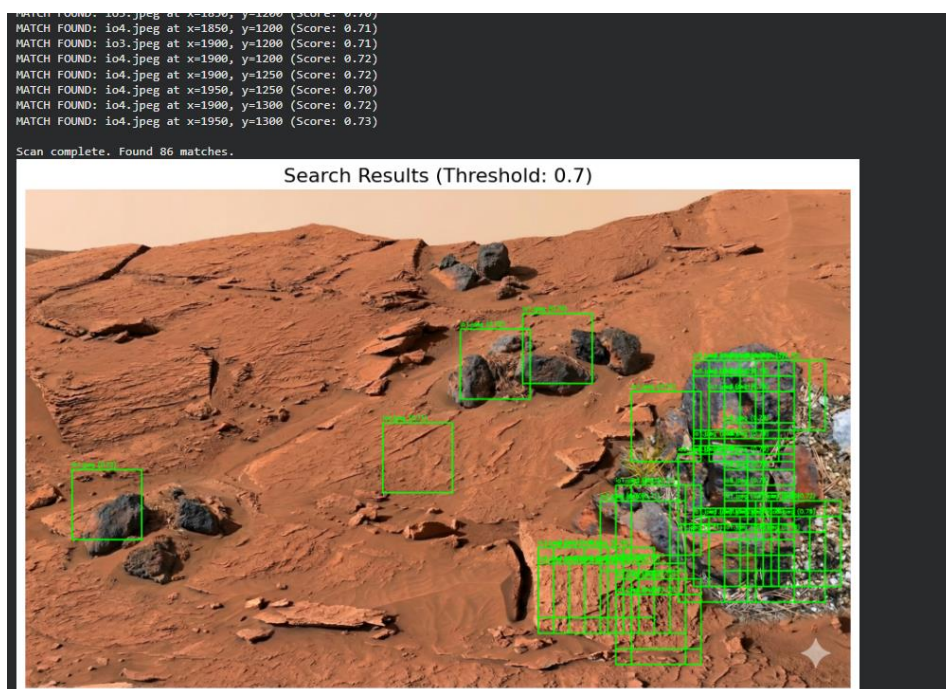
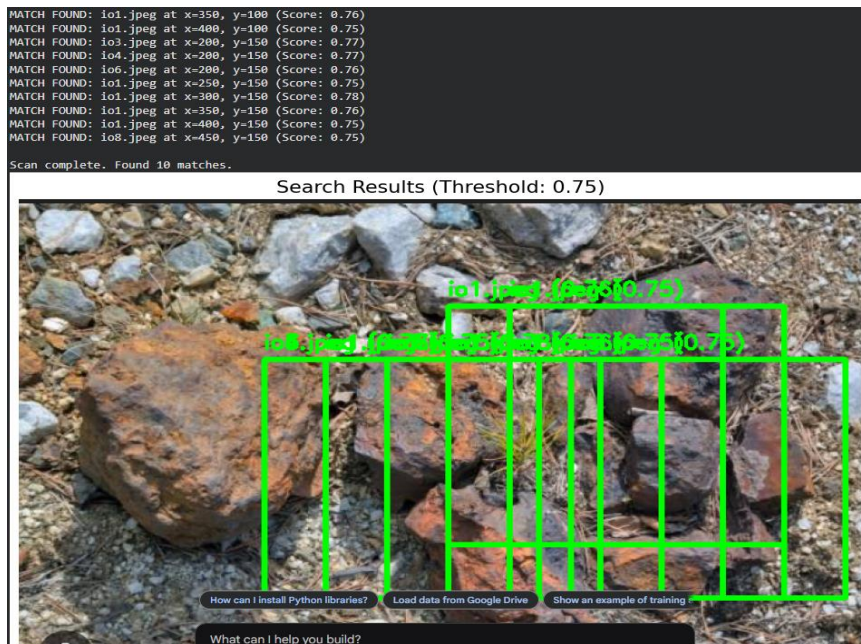


Figure 2 and 3 : Detection of Iron Oxide using ResNet-50. The model successfully identified the target texture with a confidence score of 0.75, distinguishing it from the surrounding terrain.

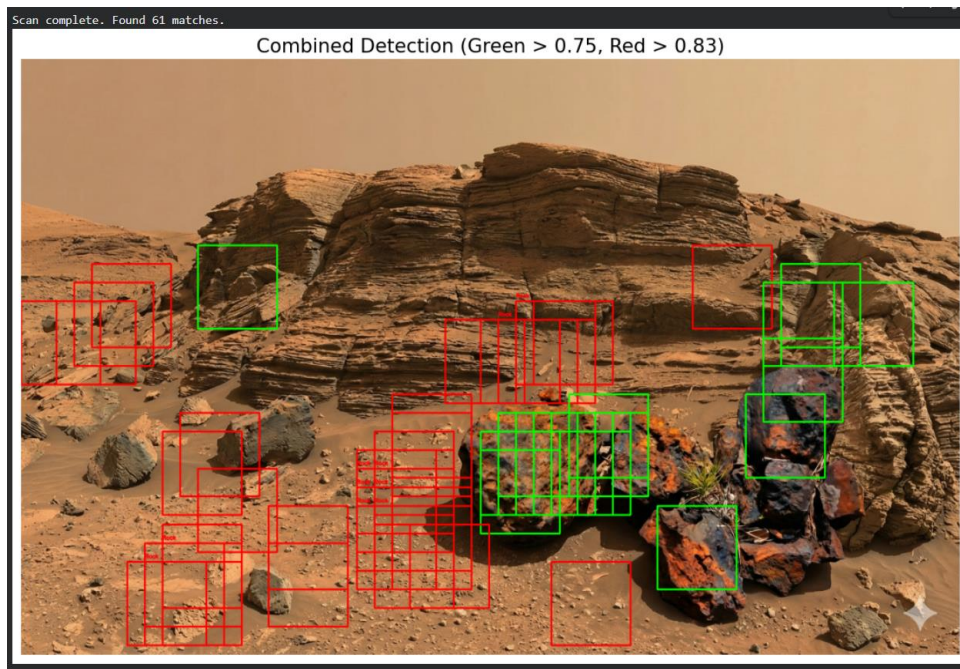


Figure 4: Simultaneous detection of Iron Oxide and Layered Rock formations using ResNet-50. The system successfully differentiated between the two classes based on feature similarity, correctly identifying Iron Oxide (green bounding boxes) and Layered Rock deposits (red bounding boxes) within the same scene."